

DATA 266 Assignment 4

Building a Mini GPT from Scratch

DATA 266 - Assignment 4 - findings document

1. Overview

A GPT style decoder only Transformer built entirely from `nn.Linear`, `nn.LayerNorm`, `nn.GELU` and `nn.Embedding`, trained on the character level Shakespeare corpus and sampled with greedy, temperature and top-k decoding. The multi-head attention, the causal mask, the decoder block and the sampling loop are all written by hand; `nn.Transformer`, `nn.MultiheadAttention` and HuggingFace models are not used.

All numbers below come from one end to end run of `notebooks/assignment4_mini_gpt.ipynb`.

2. Data preparation

The corpus is 1,115,394 characters. Tokenising at the character level gives a vocabulary of the **65** distinct characters in the file, so the mapping is exact and invertible - there is no unknown token.

Sequence length is 128. A window starting at position t gives input `[t ... t+127]` and target `[t+1 ... t+128]`, so the target is the input shifted left by one and every one of the 128 positions is a next character prediction. Windows are cut with a stride of 32, giving 34,853 training windows.

Example pair:

```
input  : 'First Citizen:\nBefore we proceed any further, hear me speak.\n\nAll:\nSpe'\n        target : 'irst Citizen:\nBefore we proceed any further, hear me speak.\n\nAll:\nSpea'
```

3. Architecture

Multi-head masked self attention. Queries, keys and values come from separate linear projections; the model dimension is split into 4 heads and all heads run in parallel through one batched matmul. Scores are scaled by the square root of the head dimension, which keeps the softmax out of its saturated regime where gradients vanish. The causal mask is applied to the *scores* with `masked_fill(..., -inf)` before the softmax, so position i attends only to positions up to i ; masking after the softmax would leave the rows no longer summing to one.

Decoder block. Attention, residual, LayerNorm, then a `Linear -> GELU -> Linear` feedforward network at four times the model width, residual, LayerNorm.

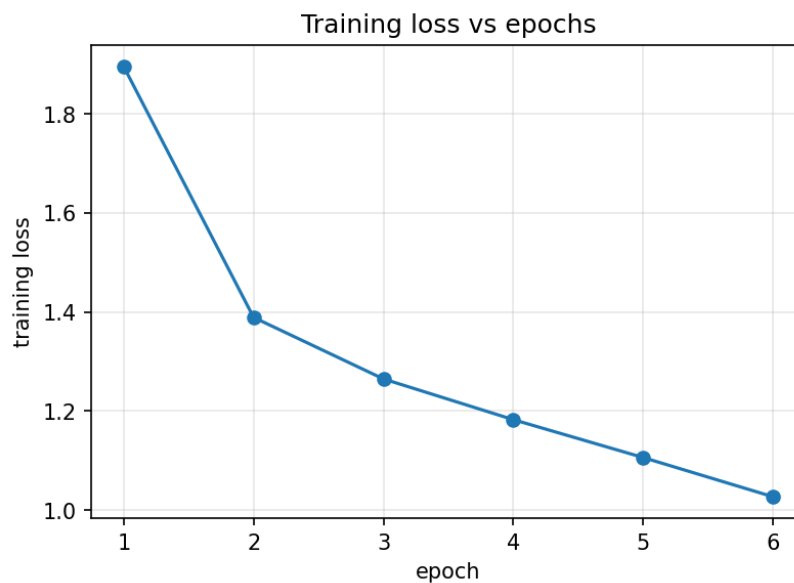
Model. Token embedding plus a learnable positional embedding, four decoder blocks, and a final linear projection to vocabulary logits. The positional embedding is required because self attention on its own is permutation equivariant.

Property	Value
Total parameter count	3,225,153
Hidden dimension	256
Number of heads	4
Number of layers	4
Context length	128
Vocabulary size	65

4. Training

Adam at a learning rate of 1e-3, batch size 64, 6 epochs.

Epoch	1	2	3	4	5	6
Training loss	1.8955	1.3892	1.2651	1.1829	1.1064	1.0272



Training loss vs epochs

The loss falls from 1.8955 to 1.0272 and is still decreasing at epoch 6, against a uniform baseline of $\ln(65) = 4.17$.

5. Decoding comparison

Each setting was scored on two measures, averaged over three generations of 300 characters from the prompt `ROME0`:

- **valid word rate** - fraction of words that occur in the corpus. A character level model must spell for itself, so this measures **coherence**.
- **distinct-3** - fraction of character trigrams that are unique. This measures **diversity**; a low value means the text repeats itself.

Decoding	Valid words (coherence)	distinct-3 (diversity)
greedy	0.984	0.671
temperature 0.5	0.984	0.753
temperature 1.0	0.864	0.864
temperature 1.5	0.755	0.905
top-k, k = 2	0.932	0.746
top-k, k = 40	0.869	0.859

Greedy

Takes the argmax at every step, so it is deterministic - the three “samples” it contributes to the average are the same string. It scores the joint best valid word rate (0.984) and by a wide margin the worst diversity (0.671, the lowest of all six settings). The failure mode is visible in the sample below: “the wisest course of the world” and “against the world” circle back to the same phrase, because with no randomness the model cannot leave a high probability region once it enters one.

```

ROMEO:
I am sorry and the wisest course of the world,
That we shall have seen them against the world.

ROMEO:
And what then it is the matter for his courtesy.

JULIET:
Ay, by this manner of my brother's head.
```

Temperature

Divides the logits by T before the softmax: T below 1 sharpens the distribution toward the most likely character, T above 1 flattens it toward uniform. The three settings are monotone in both measures - as T rises from 0.5 to 1.5 the valid word rate falls (0.984, 0.864, 0.755) while diversity rises (0.753, 0.864, 0.905). T = 0.5 stays near the mode and spells almost as well as greedy while still varying between runs. T = 1.5 flattens the distribution enough that unlikely characters get picked regularly, producing invented words such as *droans*, *sligg* and *inoriviable*, with roughly one word in four not a real one:

```

ROMEO:
Must I? Say you, sHe'll bury droans
Ppeer Dian-but the rest, or bous aniel
gone, ay eagle: Him royal can eye's for you counsel; over you have
Kather condemndly keeps, bush you and you would
Will dart you thrither sligg. Nimlips on either:
Jointice, am inoriviable; strokes--lean!---0: Mine
```

Top-k

Keeps the k most likely characters and renormalises, deleting the tail rather than reshaping the whole distribution. k = 2 is clearly more coherent than k = 40 (0.932 vs 0.869) and clearly less diverse (0.746 vs 0.859) - a gap of 0.113 in diversity, so the knob does real work in this range. At k = 2 the model is choosing between only two characters at each step and starts to lock on:

```
ROMEO:  
If they say 'nointed.' 'He killed him,'  
'Here is a charment to the sea,  
'Twas 'tis the farthe that would be done  
To see a prophecation.' Ang bold with the senation  
To be an angry angry angry and man of my brattle.
```

Worth noting: **k = 40 produced output byte identical to T = 1.0**. With only 65 characters in the vocabulary, the 25 lowest ranked ones carry too little probability to ever be sampled, so truncating them changed nothing at all. Top-k only does real work when k is small relative to the number of characters the model is actually considering - which on a sub-word vocabulary of 50,000 tokens would almost always be the case.

6. Conclusion

Most coherent: greedy. It ties with $T = 0.5$ on valid word rate (0.984 each), but it is the more conservative of the two: it only ever emits the character the model is most confident about, and its diversity score is the lowest of any setting (0.671). That is the trade - the same conservatism that keeps it spelling correctly is what makes it deterministic and repetitive, so $T = 0.5$ (0.984 coherence at 0.753 diversity) is the better practical choice when any variety is wanted.

Most diverse: temperature 1.5. It has the highest distinct-3 score (0.905), clear of every other setting. That diversity is bought with the worst coherence (0.755) - about one word in four is invented.

Decoding therefore trades coherence against diversity, and the two knobs do it differently: temperature rescales the entire distribution, whereas top-k truncates its tail and leaves the head untouched.